

z5 (5.1, 5.2)

Rozwiąż w przedziale $[0, 2\pi)$:

$$\sin^4 x + \cos^4 x = \frac{3}{4}$$

① Rozpisujemy lewą stronę równania.

$$\sin^4 x + \cos^4 x = 1 - 2\sin^2 x \cos^2 x$$

Skoro $(a+b)^2 = a^2 + 2ab + b^2$, to $a^2 + b^2 = (a+b)^2 - 2ab$.

Dla $a = \sin^2 x$ i $b = \cos^2 x$ mamy:

$$\sin^4 x + \cos^4 x = (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = 1 - 2\sin^2 x \cos^2 x$$

② Rozwiązujemy równanie.

$$1 - 2\sin^2 x \cos^2 x = \frac{3}{4}$$

$$\frac{1}{4} = 2\sin^2 x \cos^2 x$$

$$\sin^2 x \cos^2 x = \frac{1}{8}$$

$$\frac{1}{4} \sin^2(2x) = \frac{1}{8}$$

$$\sin^2(2x) = \frac{1}{2}$$

$$\sin(2x) = \frac{\sqrt{2}}{2} \quad \sin(2x) = -\frac{\sqrt{2}}{2}$$

$$2x = \frac{\pi}{4} + 2k\pi \vee 2x = \frac{3\pi}{4} + 2k\pi \vee 2x = \frac{5\pi}{4} + 2k\pi \vee 2x = \frac{7\pi}{4} + 2k\pi$$

$$x = \frac{\pi}{8} + k\pi \quad x = \frac{3\pi}{8} + k\pi \quad x = \frac{5\pi}{8} + k\pi \quad x = \frac{7\pi}{8} + k\pi, \quad k \in \mathbb{Z}$$

$$\text{Dla } k=0: x = \frac{\pi}{8}, x = \frac{3\pi}{8}, x = \frac{5\pi}{8}, x = \frac{7\pi}{8}$$

$$\text{Dla } k=1: x = \frac{9\pi}{8}, x = \frac{11\pi}{8}, x = \frac{13\pi}{8}, x = \frac{15\pi}{8}$$

$$\text{Odp.: } x \in \left\{ \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \right\}$$